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United States Patent	12415457
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Maiorana; James A.

Safety zone with breakaway for use on a tanker truck

Abstract

A vehicle includes a safety arm and a coupler on the safety arm. The coupler couples the first safety arm to the vehicle. The first safety arm is adapted to fall away from the vehicle without damaging the vehicle when the safety arm contacts an object.

Inventors:	Maiorana; James A. (Hartville, OH)
Applicant:	Mac LTT, Inc. (Kent, OH)
Family ID:	1000008822226
Assignee:	Mac LTT, Inc. (Kent, OH)
Appl. No.:	17/144508
Filed:	January 08, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210323464 A1	Oct. 21, 2021

Related U.S. Application Data

us-provisional-application US 63013255 20200421

Publication Classification

Int. Cl.: **B60Q1/30** (20060101); **B60P3/22** (20060101); **B60Q1/04** (20060101); **G08G1/16** (20060101)

U.S. Cl.:

CPC **B60Q1/305** (20130101); **B60P3/2245** (20130101); **B60P3/2265** (20130101); **B60P3/228** (20130101); **G08G1/168** (20130101); B60Q1/0491 (20130101)

Field of Classification Search

CPC: B60Q (1/305); B60P (3/2245); B60P (3/228); G08G (1/168)

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Primary Examiner: Hurley; Kevin

Attorney, Agent or Firm: Sand, Sebolt & Wernow Co., LPA

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit of U.S. Provisional Application Ser. No. 63/013,255, filed on Apr. 21, 2020, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to vehicles. More specifically, the present disclosure relates to safety systems for use with a vehicle. More specifically, the present disclosure relates to deployable safety arms for use with a vehicle.

BACKGROUND

(2) Cargo carrying vehicles, including tanker trucks, must stop at various destinations in order to unload their cargo. Typically, after arriving at a destination, a driver must exit the cab of the truck in order to begin the unloading process. During the unloading process, the driver typically stands next to the truck. Often, the unloading process is at a high traffic location, like a gas station. As a result, when the driver is at risk of being struck by another vehicle while unloading cargo. Furthermore, the unloading process may be performed at night when the driver may be less visible to other drivers. The unloading process may require the driver's full attention, making the driver unaware of potential hazards.

SUMMARY

(3) Currently, truck drivers must leave their vehicle to perform tasks that may divert their attention away from their surroundings. These tasks may be performed in high traffic areas putting the driver at risk from being struck by another vehicle. Thus, there is a continuous need for safety systems that protect drivers outside of their vehicle. In order to address this continuous need, the present disclosure provides safety arms for protecting drivers outside of their vehicle. As such, aspects of the present disclosure relate to safety arms for use with a vehicle.

(4) In one aspect, an exemplary embodiment of the present disclosure may further provide a vehicle with a first safety arm and a first coupler on the first safety arm. The first coupler may couple the first safety arm to the vehicle. The first safety arm may be adapted to fall away from the first vehicle without damaging the vehicle when the vehicle contacts an object. This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer. This embodiment or another exemplary

embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the cab. This embodiment or another exemplary embodiment may further provide a second safety arm; a second coupler on the second safety arm, wherein the second coupler couples the second safety arm to the vehicle and wherein the second safety arm is adapted to fall away from the vehicle without damaging the vehicle when the second safety arm contacts the object. This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer and the second safety arm is coupled to the cab.

(5) This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm and the second safety arm are coupled to the trailer. This embodiment or another exemplary embodiment may further provide a first side of the trailer; a second side of the trailer that is opposite the first side; and a nozzle assembly on the first side, wherein the first safety arm and the second safety arm are coupled to the first side on either side of the nozzle assembly. This embodiment or another exemplary embodiment may further provide a hose holder connected to the first side of the trailer adjacent to the nozzle assembly, wherein the first safety arm and the second safety arm are coupled to the hose holder. This embodiment or another exemplary embodiment may further provide a ferromagnetic surface, wherein the first coupler includes a magnet that couples the first safety arm to the ferromagnetic surface. This embodiment or another exemplary embodiment may further provide a receiver connected to the vehicle and coupled to the first coupler. This embodiment or another exemplary embodiment may further provide a cab; a trailer connected to the cab; a nozzle assembly on the trailer; and a hose holder connected to the trailer adjacent to the nozzle assembly, wherein the receiver is connected to the hose holder.

(6) This embodiment or another exemplary embodiment may further provide wherein the first coupler is adapted to break without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a keyhole and the first coupler includes a first portion within the keyhole and a second portion outside of the keyhole, that is adapted to break when the first safety arm contacts the object. This embodiment or another exemplary embodiment may further provide wherein the first coupler is adapted to fall away from the receiver with the first safety arm without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a triangular arm receiver and the first coupler rests within the triangular arm receiver. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a hemispherical surface and the first coupler includes a hemispherical member that abuts the hemispherical surface of the receiver. This embodiment or another exemplary embodiment may further provide wherein the first safety arm includes a light source.

(7) This embodiment or another exemplary embodiment may further provide wherein the first safety arm includes a battery in electrical communication with the light source. This embodiment or another exemplary embodiment may further provide a power source; and a power outlet in electrical communication with the power source and connected to a plug of the first safety arm, wherein the plug is in electrical communication with the light source. This embodiment or another exemplary embodiment may further provide wherein the first safety arm is coupled to the vehicle without any support structure depending from the first safety arm. In another aspect, an exemplary embodiment of the present disclosure may further provide a method for detaching a safety arm from a vehicle. The method may include attaching a safety arm to a vehicle. The method may further include applying a force to the safety arm. The method may further include detaching the safety arm from the vehicle with the applied force without damaging the vehicle. This embodiment or another exemplary embodiment may further provide illuminating an area adjacent to the vehicle with a light source on the arm while the arm is attached to the vehicle. This embodiment or another exemplary embodiment may further provide moving the vehicle; contacting an object with the arm

while the vehicle is moving, wherein the contact applies the force to the arm; and detaching the arm from the vehicle while the vehicle is moving without damaging the vehicle. This embodiment or another exemplary embodiment may further provide breaking a coupler of the arm with the applied force, wherein breaking the coupler detaches the arm from the vehicle without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the arm detaches from the vehicle by falling away from the vehicle without damaging the vehicle.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) A sample embodiment of the disclosure is set forth in the following description, is shown in the drawings and is particularly and distinctly pointed out and set forth in the appended claims. The accompanying drawings, which are fully incorporated herein and constitute a part of the specification, illustrate various examples, methods, and other example embodiments of various aspects of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one example of the boundaries. One of ordinary skill in the art will appreciate that in some examples one element may be designed as multiple elements or that multiple elements may be designed as one element. In some examples, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

(2) FIG. 1 (FIG. 1) is a side view of a vehicle with multiple safety arms in a stored position.

(3) FIG. 2 (FIG. 2) is a side view of a trailer of the vehicle depicted in FIG. 1.

(4) FIG. 3 (FIG. 3) is a cross sectional view of the trailer depicted in FIG. 2 along the line 3-3.

(5) FIG. 4 (FIG. 4) is a cross sectional view of a safety arm holder depicted in FIG. 3 along the line 4-4.

(6) FIG. 5 (FIG. 5) is a side view of the trailer depicted in FIG. 2 with two safety arms removed from the trailer.

(7) FIG. 6 (FIG. 6) is a side view of the trailer depicted in FIG. 2 with one safety arm coupled to a safety arm receiver.

(8) FIG. 7A (FIG. 7A) is an enlarged view of a safety arm and a safety arm receiver depicted in FIG. 6.

(9) FIG. 7B (FIG. 7B) is an enlarged view of the safety arm coupled to the safety arm receiver depicted in FIG. 6.

(10) FIG. 7C (FIG. 7C) is cross sectional view of the safety arm coupled to the safety arm receiver depicted in FIG. 7B along the line 7C-7C.

(11) FIG. 8 (FIG. 8) is an enlarged view of an embodiment of safety arm with a power plug coupled to a power outlet.

(12) FIG. 9 (FIG. 9) is a side view of the trailer depicted in FIG. 2 with two safety arms coupled to two safety arm receivers.

(13) FIG. 10 (FIG. 10) is an enlarged view of a safety arm depicted in FIG. 9 contacting an object and breaking away from a safety arm receiver.

(14) FIG. 11 (FIG. 11) is an enlarged view of a safety arm that is magnetized to a safety arm receiver.

(15) FIG. 12 (FIG. 12) depicts the safety arm depicted in FIG. 11 contacting an object and falling away from the safety arm receiver.

(16) FIG. 13 (FIG. 13) is an enlarged view of a safety arm in the operational position that is magnetized to the vehicle.

(17) FIG. 14 (FIG. 14) is an enlarged view of an embodiment of a safety arm a triangularly shaped coupler and a safety arm receiver.

- (18) FIG. 15 (FIG. 15) is an enlarged view of the safety arm depicted in FIG. 14 in coupled to a safety arm receiver.
- (19) FIG. 16 (FIG. 16) is a top view of the safety arm depicted in FIG. 15 along the line 16-16 coupled to a safety arm receiver.
- (20) FIG. 17 (FIG. 17) is a top view of the safety arm depicted in FIG. 15 along the line 16-16 contacting an object.
- (21) FIG. 18 (FIG. 18) is a side view of an embodiment of a safety arm with a ball lock and a safety arm receiver.
- (22) FIG. 19 (FIG. 19) is a side view of the safety arm depicted in FIG. 18 in the coupled to a safety arm receiver.
- (23) FIG. 20 (FIG. 20) is a top view of the safety arm coupled to a safety arm receiver as depicted in FIG. 19 along the line 20-20.
- (24) FIG. 21 (FIG. 21) is a top view of the safety arm and safety arm receiver depicted in FIG. 19 along the line 20-20 contacting an object.
- (25) FIG. 22 (FIG. 22) is a side view of a trailer with members of a safety device removed from the trailer.
- (26) FIG. 23 (FIG. 23) is a side view of a bracket of a safety device.
- (27) FIG. 24 (FIG. 24) is a side view of the bracket depicted in FIG. 23 that has been inverted.
- (28) FIG. 25A (FIG. 25A) is a side view of the bracket depicted in FIG. 23 coupled to a safety arm.
- (29) FIG. 25B (FIG. 25B) is a side view of the bracket and safety arm depicted in FIG. 25A wherein the safety arm has been rotated.
- (30) FIG. 25C (FIG. 25C) is a side view of the bracket depicted in FIG. 23 coupled to two safety arms.
- (31) FIG. 26 (FIG. 26) depicts a safety device formed by the bracket and safety arms depicted in FIG. 25C standing vertically on a surface.
- (32) FIG. 27 (FIG. 27) is a side view of the safety device depicted in FIG. 26 adjacent to a vehicle.
- (33) Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

- (34) FIG. 1 depicts a vehicle 100. While the vehicle 100 is depicted as a truck, it is envisioned the vehicle 100 may be any vehicle. The vehicle 100 may include a tractor unit or a cab 102 and a trailer 104. The cab 102 may be connected to the trailer 104 to provide a pulling force for moving the vehicle 100.
- (35) The cab 102 includes one or more pairs of wheel assemblies 106 and a power source 108. The one or more wheel assemblies 106 may be adapted to support the cab 102 and the trailer 104. The power source 108 may be carried by the trailer 104.
- (36) The trailer 104 includes a generally cylindrical tank 110, one or more pairs of wheel assemblies 112, a landing gear 114, and a nozzle assembly 116. The one or more wheel assemblies 112 may be adapted to support the tank 110. Furthermore, when the vehicle 100 is stationary, the landing gear 114 may be adapted to provide further support to the tank 110. The tank 110 includes a hollow interior for transporting material. The shape and size of the tank 110 will be dependent upon the material being transported. The nozzle assembly 116 may be on a side of the tank 110 and may be in open communication with the hollow interior of the tank 110. Hoses may be selectively engaged with the nozzle assembly 116 to transfer the material (i.e., fuel, liquid, grain, etc.) from the hollow interior of the tank 110 to a storage container. Although the trailer 104 is referred to herein as including a cylindrical tank 110, it is envisioned that the trailer 104 may include non-cylindrical trailers (i.e., rectangular cargo trailers). The trailer 104 further includes a first hose holder 118 connected to the tank 110. As shown in FIG. 2, the first hose holder 118 is adapted to hold one or more hoses 202. As further shown in FIG. 2, the trailer 104 includes a second hose holder 204 connected to the tank 110 and adapted to hold one or more hoses 202.
- (37) Returning to FIG. 1, the trailer 104 further includes and carries a safety arm holder 120 and

one or more safety arm receivers. As will be discussed in further detail below, in one embodiment, the safety arm holder **120** is connected to the first hose holder **118** and the second hose holder **204**. In another embodiment, the safety arm holder **120** may be mounted anywhere on any outer surface of the vehicle **100** (i.e., adjacent to the nozzle assembly **116**). The safety arm holder **120** may be adapted to retain one or more safety arms **124** outside of the vehicle **100** while the vehicle **100** is stationary or moving. While the safety arm holder **120** is shown holding four safety arms **124**, it is envisioned that the safety arm holder **120** may hold any number of safety arms **124** (i.e., 1, 3, 6, 9, 10, etc.). In one embodiment (FIG. 1), the safety arm holder **120** may be in electrical communication with the power source **108**. Although the power source **108** is shown within cab **102**, it is envisioned that the power source **108** may be located elsewhere on the vehicle **100**. It is further envisioned that the power source **108** may be external from the vehicle **100**.

(38) As will be discussed in further detail below, the safety arm receivers are adapted to couple to a safety arm **124** such that, when coupled, a safety arm receiver provides vertical and horizontal support for a safety arm **124** so that a safety arm **124** may extend from the vehicle **100** without any support structure depending from the safety arm **124**.

(39) The trailer **104** further includes a bracket holder **126**. Although the bracket holder **126** is shown connected to the first hose holder **118**, it is envisioned that the bracket holder **126** may be connected to the second hose holder **204** or the tank **110**. Furthermore, the bracket holder **126** may be adapted to contain a bracket **128** and a bracket support **130**. As will be discussed in further detail below, the bracket **128** may be adapted to receive one or more safety arms **124** and the bracket support **130**. The bracket support **130** may be adapted to support the bracket **128**.

(40) FIG. 2 depicts the tank **110**. The tank **110** includes a first end **206**, a second end **208**, a first side **210**, a second side **212**, a top side **214**, and a bottom side **216**. The first end **206** faces the cab **102** and the second end **208** is opposite the first end **206**. A central longitudinal axis **218** extends between and through the first end **206** and the second end **208** thereby defining a longitudinal direction between the first end **206** and the second end **208**. The first side **210** is opposite the second side **212** and a central transverse axis **220** extends between and through the first side **210** and the second side **212** thereby defining a transverse direction between the first side **210** and the second side **212**. The top side **214** is opposite the bottom side **216** and a central vertical axis **222** extends between and through the top side **214** and the bottom side **216** thereby defining a vertical direction between the top side **214** and the bottom side **216**. The central longitudinal axis **218**, the central transverse axis **220**, and the central vertical axis **222** are oriented at right angles relative to one another.

(41) FIG. 3 depicts safety arms **124** in a stored position. The safety arm holder **120** slidably receives and supports the safety arms **124** and the safety arms **124** extend parallel to the central transverse axis **220** and perpendicular to the central longitudinal axis **218**.

(42) The safety arm holder **120** includes a first holder **302** a distance away from a second holder **304**. The distance between the first holder **302** and the second holder **304** is less than a length of a safety arm **124**. As such, a safety arm **124** may be carried and retained by the first holder **302** and the second holder **304** when the vehicle **100** is stationary or moving.

(43) In one embodiment, the first holder **302** may be connected to the first side **210** of the tank **110** and the second holder **304** may be connected to the second side **212** of the tank **110**. In this embodiment, the first holder **302** may be connected to a bottom surface **306** of the first hose holder **118** and the second holder **304** may be connected to a bottom surface **308** of the second hose holder **204**. The first holder **302** and the second holder **304** are further connected to a mounting member **310** and **312** respectively. The mounting members **310** and **312** are connected to the tank **110**. In another embodiment, the first holder **302** and the second holder **304** may be anywhere on the vehicle **100** (i.e., on the first side **210** of the tank **110**, on the second side **212** of the tank **110**, etc.).

(44) The first holder **302** includes a first end **314**, a second end **316** opposite the first end **314**, a top end **318**, and a bottom end **320** opposite the top end **318**. The first end **314** extends longitudinally

relative to the central vertical axis **222** from the bottom surface **306** to the bottom end **320** and the second end **316** extends longitudinally relative to the central vertical axis **222** from the mounting member **310** to the bottom end **320**. Opposing inner surfaces **322** extend longitudinally relative to the central vertical axis **222** from the top end **318** to the bottom end **320** and longitudinally relative to the central longitudinal axis **218** from the first end **314** to the second end **316**. The first end **314** and the second end **316** are open ended such that when a safety arm **124** is moved in either direction of arrow **A**, a safety arm **124** may pass through the first holder **302**.

(45) The first holder **302** further includes a first rack **324** and a second rack **326**. The first rack **324** is vertically above second rack **326**. Stated another way, the second rack **326** is vertically below the first rack **324**. The first rack **324** and the second rack **326** extend between the first end **314** and the second end **316** of the first holder **302**. Furthermore, the first rack **324** and the second rack **326** are adapted to slidably receive and support one or more safety arms **124**.

(46) The second holder **304** includes a first end **328**, a second end **330** opposite the first end **328**, a top end **332**, and a bottom end **334** opposite the top end **332**. The first end **328** extends longitudinally relative to the central vertical axis **222** from the bottom surface **308** to the bottom end **334** and the second end **330** extends longitudinally relative to the central vertical axis **222** from the mounting member **312** to the bottom end **334**. Opposing inner surfaces **336** extend longitudinally relative to the central vertical axis **222** from the top end **332** to the bottom end **334** and longitudinally relative to the central vertical axis **222** from the first end **328** to the second end **330**.

(47) The second holder **304** further includes a first rack **338** and a second rack **340**. The first rack **338** is vertically above second rack **340**. Stated another way, the second rack **340** is vertically below the first rack **338**. The first rack **338** and the second rack **340** extend between the first end **328** and the second end **330** of the second holder **304**. The first rack **338** and the second rack **340** are adapted to slidably receive and support one or more safety arms **124**.

(48) In one embodiment only the second end **330** of the second holder **304** is open ended. In this embodiment, when placing a safety arm **124** into the stored position, a safety arm **124** enters the safety arm holder **120** from the first side **210** of the tank **110**, passes through the first end **314** of the first holder **302** and through the second end **330** of the second holder **304**. Stated another way, a safety arm **124** first enters the first holder **302** then enters the second holder **304**.

(49) In another embodiment, the first end **328** and the second end **330** of the second holder **304** are open ended. In this embodiment, when placing a safety arm **124** in a stored position, a safety arm **124** may enter the safety arm holder **120** from the first side **210** or the second side **212** of the tank **110**. Stated another way, a safety arm **124** may first pass through the first holder **302** and then into the second holder **304** or may first pass through the second holder **304** and then into the first holder **302**.

(50) In one embodiment (FIG. 3), the second holder **304** may be in electrical communication with the power source **108**. In this embodiment, the second holder **304** includes a charging dock **342** and a safety arm **124** includes and carries a battery **344**. The charging dock **342** is carried by the trailer **104** and is in electrical communication with the power source **108** and is adapted to charge the battery **344** of a safety arm **124** when a safety arm **124** is connected to the charging dock **342**. In this embodiment, the battery **344** is in electrical communication and supplies power to a light source **345** of a safety arm **124**.

(51) The charging dock **342** includes a number of receptacles or channels **346**. The number of channels **346** may depend upon the number of safety arms **124** that the safety arm holder **120** is adapted to retain. For example, in the embodiment shown in FIG. 2, the safety arm holder **120** is adapted to retain four safety arms **124**. Hence, the charging dock **342** may contain four channels **346**. It is envisioned that the charging dock **342** may contain any number of channels **346** (i.e., 1, 3, 6, 9, 10, etc.). The channels **346** are defined by a surface **348**. The surface **348** extends from a side surface **352** to an opening **354**. The surface **348** defines a shape of a channel **346**. The shape of a

channel **346** may correspond to a shape and a dimension of a safety arm **124**. For example, in the embodiment shown in FIG. **4**, a safety arm **124** has a circular shape. As such, a channel **346** may be circular with a diameter that corresponds to a diameter of a safety arm **124**. Due to the corresponding shape and dimension, when a safety arm **124** is in the stored position, a portion of the safety arm **124** nests within the channel **346**.

(52) In the embodiment shown in FIG. **3**, a channel **346** further includes a charging connector **358**. The charging connector **358** is in electrical communication with the power source **108**.

Furthermore, a safety arm **124** includes a first end **360**, a second end **362** opposite the first end **360** defining a longitudinal direction there between. In the embodiment shown in FIG. **3**, a charging receptacle **364** extends inward from the second end **362** towards the first end **360** and is in electrical communication with the battery **344**. The charging receptacle **364** has a complementary shape to the charging connector **358** and is adapted to slidably receive the charging connector **358** thereby connecting the charging connector **358** to the charging receptacle **364**. When a safety arm **124** is in the stored position, the second end **362** of a safety arm **124** abuts the side surface **352** of the channel **346** and the charging receptacle **364** connects to the charging connector **358**. The complementary shapes of the charging connector **358** and the charging receptacle **364** provides a frictional interference fit between the charging connector **358** and the charging receptacle **364**.

When the charging connector **358** is connected to the charging receptacle **364**, the battery **344** is in electrical communication with the power source **108** which allows the power source **108** to charge the battery **344**. The charging dock **342** may charge any number of safety arms **124** simultaneously.

(53) As will be discussed in further detail below, a safety arm **124** further includes coupler that is adapted to attach or couple a safety arm **124** to the vehicle **100**. A safety arm **124** is adapted to fall away from the vehicle **100** without damaging the vehicle **100** (i.e., the cab **102** or the trailer **104**).

(54) FIG. **4** depicts the second holder **304**. As shown in FIG. **4**, the first rack **338** and the second rack **340** of the second holder **304** are structurally identical. For the sake of brevity, reference numerals on the first rack **338** correspond to identical elements on the second rack **340**.

Furthermore, it is understood that the first rack **338** and the second rack **340** are structurally identical to the first rack **324** and the second rack **326** of the first holder **302**. The first rack **338** includes a first surface **402** and a second surface **404** opposite the first surface **402**. The first surface **402** includes a number of grooves **406**. The number of grooves **406** will be dependent upon a number of safety arms **124** that the safety arm holder **120** is adapted to retain. For example, in the embodiment shown in FIG. **2**, the safety arm holder **120** is adapted to retain four safety arms **124**. Hence, the first rack **338** contains two grooves **406** and the second rack **340** contains two grooves **406** for a total of four grooves **406**. It is envisioned that either the first rack **338** or the second rack **340** may contain any number of grooves **406** (i.e., 1, 3, 6, 9, 10, etc.). It is further envisioned that the first holder **302** and the second holder **304** may contain any number of racks (1, 3, 6, 9, 10, etc.).

(55) Furthermore, a shape and dimension of a groove **406** corresponds to a shape and dimension of a safety arm **124**. For example, in the embodiment shown in FIG. **4**, a groove **406** is semi-circular. The semi-circular shape corresponds to the circular shape and dimension of the safety arm **124**. Due to the corresponding shape and dimension, when a safety arm **124** is in the stored position, the safety arm **124** nests within the groove **406**. When the second holder **304** includes a charging dock **342**, the surface **348** of a channel **346** is flush with groove **406** which allows a safety arm **124** to slide into the channel **346** from a first rack **338** or a second rack **340**.

(56) FIG. **5** depicts safety arms **124** removed from the trailer **104**. In order to remove safety arms **124** from the trailer **104**, safety arms **124** may be slidably moved in the direction of arrow B.

(57) FIG. **6** depicts a breakaway assembly **600**. In this embodiment the breakaway assembly **600** includes a coupler **602** at the first end **360** of a safety arm **124** and a safety arm receiver **604** on the vehicle **100**.

(58) In this embodiment, the coupler **602** couples to safety arm receiver **604** by moving the safety

arm **124** in the direction of arrow C. The safety arm receiver **604** includes a keyhole **606** with a first portion **608** and a second portion **610**. The keyhole **606** receives the coupler **602** of a safety arm **124**.

(59) FIGS. 7A-7C depict a safety arm **124** with coupler **602** and the safety arm receiver **604**. The safety arm receiver **604** includes a first wall **605** and a second wall **607** that extend from the bottom surface **306** of the first hose holder **118**. The second wall **607** includes an outer surface **612** and an inner surface **614**. The keyhole **606** extends from the outer surface **612** to the inner surface **614**. Although the safety arm receiver **604** is shown connected to the first hose holder **118**, it is envisioned that the safety arm receiver **604**, and other embodiments of arm receivers, may be connected to the second hose holder **204**, the tank **110**, or the cab **102**.

(60) The coupler **602** includes a first circular wall **620**. The first circular wall **620** includes a first circular surface **622** and a second circular surface **624** opposite the first circular surface **622**. The coupler **602** further includes a second circular wall **626** that extends from the second circular surface **624** to a circular surface **628** of a first tapered portion **630** of the coupler **602**. The first circular wall **620** has a greater diameter than the second circular wall **626**. The first tapered portion **630** extends from the circular surface **628** to a break portion **632** of the coupler **602**. The break portion **632** extends from the first tapered portion **630** to a second tapered portion **634**.

(61) FIG. 7B depicts a safety arm **124** coupled to safety arm receiver **604**. The keyhole **606** is dimensioned so that the first circular wall **620** may pass through the first portion **608** of the keyhole **606**. The first circular wall **620** may pass through the first portion **608** by moving a safety arm **124** in the direction of arrow D. When the first circular wall **620** passes through the first portion **608** and the second circular wall **626** aligns with the second portion **610** of the keyhole **606**, the safety arm **124** is then moved in the direction of arrow E thereby coupling the coupler **602** to the safety arm receiver **604**.

(62) FIG. 7C further depicts a safety arm **124** coupled to a safety arm receiver **604**. The keyhole **606** extends from the outer surface **612** of the first wall **605** of the safety arm receiver **604** to the inner surface **614** of the first wall **605**. When the coupler **602** is couple to the safety arm receiver **604**, the second circular surface **624** abuts the inner surface **614** of the first wall **605**, the circular surface **628** abuts the outer surface **612** of the first wall **605** and the second circular wall **626** nests within the second portion **610** of the keyhole **606**.

(63) FIG. 8 depicts an embodiment wherein a power outlet **802** is connected to the first hose holder **118** and adjacent to the safety arm receiver **604**. In another embodiment, the power outlet **802** may be connected to the second hose holder **204** or anywhere else on the vehicle **100**. The power outlet **802** is in electrical communication with the power source **108**. In this embodiment, the power outlet **802** is adapted to receive a plug **804** of a safety arm **124**. The plug **804** is connected to a wire **806** that places the plug **804** in electrical communication with the light source **345**. As such, when the power outlet **802** receives the plug **804**, the power source **108** supplies power to the light source **345** of the safety arm **124**.

(64) FIG. 9 depicts two safety arms **124** in an operational position. When in the operational position, one safety arm **124** or two safety arms **124**, as depicted in FIG. 9, extend outwardly from the vehicle **100** thereby creating a safety zone. While FIG. 9 depicts two safety arms **124** coupled to the trailer **104**, it is understood that any number of safety arms **124** may be coupled to the vehicle **100** (i.e., the trailer **104** or the cab **102**). The safety arms **124** include a first side **902**, a second side **904** opposite the first side **902**, and power switch **906**. While FIG. 9 depicts the light source **345** on the first side **902** and the second side **904** of the safety arm **124**, it is envisioned that the safety arm **124** may include the light source **345** on only the first side **902** or only the second side **904**. In this embodiment, the two safety arms **124** are coupled to the first side **210** and adjacent to the nozzle assembly **116**.

(65) The power switch **906** controls the flow of electricity from a battery **344** or the power source **108** to the light source **345**. A user may engage the power switch **906** a number of times to power a

light source **345**. For example, when a safety arm **124** includes the light source **345** on the first side **902** and the second side **904** and is off (i.e., no light is being emitted from the light source **345** on either the first side **902** or the second side **904**), a user may engage the power switch **906** a first time to cause a light source **345** to emit light from the first side **902** and the second side **904**. After engaging the power switch **906** a first time to emit light from both the first side **902** and the second side **904**, the user may engage the power switch **906** a second time to emit light from only one side (i.e., the first side **902** or the second side **904**) of the safety arm **124**. The user may then engage the power switch **906** a third time to power off the safety arm **124** (i.e., no light is being emitted from the light source **345** on either the first side **902** or the second side **904**). When on, the light source **345** emits light **908** that illuminates an area adjacent to the vehicle **100**. The light source **345** may be any suitable source of illumination including, but not limited to, at least one strip of light emitting diodes.

(66) FIG. **10** depicts a safety arm **124** contacting an object **1002** (i.e., a stationary pole). In one embodiment, the coupler **602** may be formed from a suitably rigid material so as to support the safety arm **124** when the safety arm **124** is coupled to the safety arm receiver **604** without any support structure depending from the safety arm **124**. Furthermore, the integral structure of the coupler **602** may be formed from multiple elements having similar configurations as one having ordinary skill in the art would understand.

(67) In one embodiment, the first tapered portion **630**, the break portion **632**, and the second tapered portion **634** are formed of the same material. In another embodiment, the first tapered portion **630**, the break portion **632**, and the second tapered portion **634** are formed of different materials. In this embodiment, the break portion **632** is formed of a material that is less rigid, and therefore more susceptible to breaking, than a material that forms a remainder of the coupler **602** including the first tapered portion **630** and the second tapered portion **634**. Furthermore, the break portion **632** has a smaller diameter than the first tapered portion **630** and the second tapered portion **634**. As such, the break portion **632** is less rigid and more susceptible to breaking than the remainder of the coupler **602** including first tapered portion **630** or the second tapered portion **634**.

(68) In one example, when the vehicle **100** moves in the direction of arrow F and the object **1002** contacts the safety arm **124**, a force is applied to the safety arm **124** causing the safety arm **124** to move in the direction of arrow G. This force may cause the coupler **602** to break within the break portion **632**. Due to the location of the break portion **632**, when the break portion **632** breaks, the vehicle **100** is undamaged. When the break portion **632** breaks, the safety arm **124** falls away from the vehicle **100**.

(69) FIG. **11** depicts a breakaway assembly **1100**. In this embodiment, the breakaway assembly **1100** includes a coupler **1102** at the first end **360** of a safety arm **124** and a safety arm receiver **1104** on the vehicle **100**.

(70) In this embodiment, the coupler **1102** includes a magnet that couples to a safety arm receiver **1104** that is formed from a suitable ferromagnetic material. The safety arm receiver **1104** includes a first wall **1106** that includes a surface **1108**. The coupler **1102** includes a surface **1110** that defines the first end **360** of a safety arm **124** and magnetizes the coupler **1102** to the surface **1108** of the safety arm receiver **1104**. In this embodiment, the coupler **1102** may be suitably strong to support the safety arm **124** when the safety arm **124** is coupled to the safety arm receiver **1104** without any support structure depending from the safety arm **124**.

(71) FIG. **12** depicts a safety arm **124** contacting the object **1002**. In this embodiment, the coupler **1102** may not be strong enough to support the safety arm **124** when the safety arm **124** contacts the object **1002**. For example, when the vehicle **100** moves in the direction of arrow H and the object **1002** contacts the safety arm **124**, a force is applied to the safety arm **124** causing the safety arm **124** to move in the direction of arrow I. This force causes the coupler **1102** to demagnetize from the safety arm receiver **1104**. When the coupler **1102** demagnetizes from the safety arm receiver **1104** due to a force applied to the safety arm **124**, the safety arm **124** falls away from the vehicle **100**.

with the coupler **1102** and without damaging the vehicle **100**. In this embodiment, the safety arm **124** may be undamaged and the safety arm **124** may be recoupled to the safety arm receiver **1104**. (72) Although FIGS. **11** and **12** depict the coupler **1102** coupled to a safety arm receiver **1104**, it is envisioned that the coupler **1102** may couple a safety arm **124** to any ferromagnetic surface of the vehicle **100**. For example, FIG. **13** depicts a breakaway assembly **1300**. In this embodiment, the breakaway assembly **1300** includes the coupler **1102** and a safety arm receiver **1304**. In this embodiment, the safety arm receiver **1304** may be a disk extending from a surface **118A** of the first hose holder **118**. While the safety arm receiver **1304** is depicted as extending from the first hose holder **118**, it is envisioned that the safety arm receiver **1304** may extend any surface of the vehicle **100**. The safety arm receiver **1304** may be made formed of a ferromagnetic material or a magnet that attracts the coupler **1102**.

(73) FIG. **14** depicts a breakaway assembly **1400**. In this embodiment, the breakaway assembly **1400** includes a coupler **1402** and a safety arm receiver **1404**. The coupler **1402** is generally triangular in shape and the safety arm receiver **1404** has a complimentary triangular shape.

(74) In this embodiment, the coupler **1402** defines the first end **360** of a safety arm **124**. A neck **1406** extends from the safety arm **124** to a first surface **1408** of the coupler **1402**. The first surface **1408** extends between a second surface **1410** and a third surface **1412**. A fourth surface **1414** and a fifth surface **1416** opposite the fourth surface **1414** extend from and generally perpendicular to the first surface **1408** to a first angled surface **1418** and a second angled surface **1420**. The fourth surface **1414** and the fifth surface **1416** extend between the second surface **1410** and the third surface **1412**. The first angled surface **1418** is opposite the second angled surface **1420**. The first angled surface **1418** and the second angled surface **1420** extend from the fourth surface **1414** and the fifth surface **1416** respectively to a rounded corner **1422**. Furthermore, the first angled surface **1418**, the second angled surface **1420**, and the rounded corner **1422** extend between the second surface **1410** the third surface **1412**.

(75) The safety arm receiver **1404** is adapted to receive the coupler **1402** and includes a bottom wall **1405** and a top wall **1407**. The safety arm receiver **1404** further includes a first outer edge **1423**, a second outer edge **1424** opposite the first outer edge **1423**, a third outer edge **1426**, and a fourth outer edge **1428** opposite the third outer edge **1426**. The third outer edge **1426** and the fourth outer edge **1428** extend between the bottom wall **1405** and the top wall **1407**. The first outer edge **1423** and the second outer edge **1424** extend between and generally perpendicular to the third outer edge **1426** and the fourth outer edge **1428**. The third outer edge **1426** and the fourth outer edge **1428** extend between and generally perpendicular to the first outer edge **1423** and the second outer edge **1424**. The first outer edge **1423**, the second outer edge **1424**, the third outer edge **1426**, and the fourth outer edge **1428** define an aperture **1411** that receives the coupler **1402**.

(76) The safety arm receiver **1404** further includes an angled wall **1409**. The angled wall **1409** includes a first angled surface **1430**, a second angled surface **1432** opposite the first angled surface **1430**, a rounded corner **1434**. The safety arm receiver **1404** further includes a first surface **1436**, and a second surface **1438**. The first angled surface **1430** and the second angled surface **1432** extend from the third outer edge **1426** and the fourth outer edge **1428** respectively to the rounded corner **1434**. The first angled surface **1430**, the second angled surface **1432**, and the rounded corner **1434** extend between the first surface **1436** and the second surface **1438**. The first surface **1436** and the second surface **1438** extend between the first angled surface **1430** and the second angled surface **1432**. Furthermore, the first surface **1436** extends between the first outer edge **1423** and the rounded corner **1434** and the second surface **1438** extends between the second outer edge **1424** and the rounded corner **1434**.

(77) As depicted in FIGS. **15** and **16**, when the coupler **1402** is coupled to the safety arm receiver **1404**, the coupler **1402** nests within the safety arm receiver **1404**. When the coupler **1402** nests within the safety arm receiver **1404**, the second surface **1410** of the coupler **1402** abuts the first surface **1436** of the safety arm receiver **1404**, the third surface **1412** of the of coupler **1402** abuts

the second surface **1438** of the safety arm receiver **1404**, the rounded corner **1422** of the coupler **1402** abuts the rounded corner **1434** of the safety arm receiver **1404**, the first angled surface **1418** of coupler **1402** abuts the first angled surface **1430** of the safety arm receiver **1404**, and the second angled surface **1420** of coupler **1402** abuts the second angled surface **1432** of the safety arm receiver **1404**.

(78) As depicted in FIG. **16**, the first angled surface **1418** is not as long as the first angled surface **1430** and the second angled surface **1420** is not as long as the second angled surface **1432**. As a result, the fourth surface **1414** and the fifth surface **1416** of coupler **1402** do not abut a surface of the safety arm receiver **1404**.

(79) FIG. **17** depicts a safety arm **124** contacting the object **1002**. In this embodiment, when the safety arm **124** contacts the object **1002** the safety arm **124** falls away from the vehicle **100** with coupler **1402**. For example, when the vehicle **100** moves in the direction of arrow J and the object **1002** contacts the safety arm **124**, a force is applied to the safety arm **124** causing the safety arm **124** to move in the direction of arrow K. This force causes coupler **1402** to detach or decouple from the safety arm receiver **1404**. When coupler **1402** decouples from the safety arm receiver **1404** due to a force applied to the safety arm **124**, the safety arm **124** falls away from the vehicle **100** without damaging the vehicle **100**. In this embodiment, the safety arm **124** may be undamaged and the coupler **1402** may be recoupled to the safety arm receiver **1404**.

(80) FIG. **18** depicts a breakaway assembly **1800**. In this embodiment, the breakaway assembly **1800** includes a coupler **1802** at the first end **360** of a safety arm **124** and a safety arm receiver **1803** on the vehicle **100**. While FIG. **18** depicts the safety arm receiver **1803** on the first hose holder **118**, it is envisioned that the safety arm receiver **1803** may be coupled to the second hose holder **204**.

(81) The coupler **1802** includes an extension **1805** that extends from the safety arm **124**. The coupler **1802** further includes a hook **1807** that extends from the extension **1805**. The hook **1807** includes a first leg **1809** and a second leg **1811** that is perpendicular to the first leg **1809**. The coupler **1802** further includes a first surface **1804** of the coupler **1802** extends longitudinally relative to the central vertical axis **222** from a first surface **1806** of the extension **1805** to a first rounded corner **1808**.

(82) The first rounded corner **1808** convexly curves downward relative to the central vertical axis **222** approximately 90° from the first surface **1804** to a second surface **1810** of the coupler **1802**. The second surface **1810** extends longitudinally relative to the central transverse axis **220** from the first rounded corner **1808** to a third surface **1812** of the coupler **1802**. The third surface **1812** extends longitudinally relative to the central vertical axis **222** from the second surface **1810** to a fourth surface **1814**. The fourth surface **1814** extends longitudinally relative to the central transverse axis **220** from the third surface **1812** to a fifth surface **1816** of the coupler **1802**. The fifth surface **1816** extends longitudinally relative to the central vertical axis **222** from the fourth surface **1814** to a sixth surface **1818**. The sixth surface **1818** extends longitudinally relative to the central transverse axis **220** from the fifth surface **1816** to a seventh surface **1820**. The seventh surface **1820** extends longitudinally relative to the central vertical axis **222** from the sixth surface **1818** to an eighth surface **1822**. The eighth surface **1822** extends longitudinally relative to the central transverse axis **220** from the seventh surface **1820** to a second rounded corner **1824**. The second rounded corner **1824** concavely curves downward relative to the central vertical axis **222** from the eighth surface **1822** to a terminal surface **1826** of the safety arm **124**.

(83) The first surface **1804**, the third surface **1812**, the fifth surface **1816**, the seventh surface **1820**, and the terminal surface **1826** extend generally parallel relative to the central vertical axis **222**. The second surface **1810**, the fourth surface **1814**, the sixth surface **1818**, and the eighth surface **1822** extend generally perpendicular relative to the central vertical axis **222**.

(84) A hemispherical member **1828** extends longitudinally relative to the central vertical axis from the eighth surface **1822**. As will be discussed in further detail below, the hemispherical member

1828 is adapted to nest within the safety arm receiver **1803**. The third surface **1812**, the fourth surface **1814**, the fifth surface **1816**, the sixth surface **1818**, and the seventh surface **1820** define an opening **1830**. As will be discussed in further detail below, when the coupler **1802** is coupled to the safety arm receiver **1803**, a portion of the safety arm receiver **1803** nests within the opening **1830**. (85) The safety arm receiver **1803** includes a post member **1832** and a hemispherical receptacle **1834**. The post member **1832** includes a first leg **1831** and a second leg **1833** that is perpendicular to the first leg **1831**. The safety arm receiver **1803** further includes a first wall **1836**. The first wall **1836** extends longitudinally relative to the central vertical axis **222** from the bottom surface **306** of the first hose holder **118**. The first wall **1836** includes a surface **1840** that extends longitudinally relative to the central vertical axis **222** from the bottom surface **306** to a second wall **1842**. The first wall **1836** extends generally parallel to the central vertical axis **222** and the second wall **1842** extends generally perpendicular to the central vertical axis **222**. The first wall **1836**, the second wall **1842**, and the bottom surface **306** define an inlet **1844**. As will be discussed in further detail below, in order to couple the coupler **1802** to the safety arm receiver **1803**, the coupler **1802** passes through the inlet **1844**.

(86) The post member **1832** extends longitudinally relative to the central vertical axis **222** from a surface **1846** of the second wall **1842**. The post member **1832** includes a first surface **1848**. The first surface **1848** extends longitudinally relative to the central vertical axis **222** to a second surface **1850**. The second surface **1850** extends longitudinally relative to the central transverse axis **220** to a rounded surface **1852**. The rounded surface **1852** concavely curves upward relative to the central vertical axis **222** approximately 180° from the second surface **1850** to a third surface **1854**. The third surface **1854** extends longitudinally relative to the central transverse axis **220** to a fourth surface **1856**. The fourth surface **1856** extends longitudinally relative to the central vertical axis **222** to the surface **1846** of the second wall **1842**. The first surface **1848** and the third surface **1854** extend generally parallel to the central vertical axis **222** while the second surface **1850** and the fourth surface **1856** extend generally perpendicular to the central vertical axis **222**.

(87) The hemispherical receptacle **1834** includes a first surface **1858** that extends longitudinally relative to the central vertical axis **222** from the surface **1846** of the second wall **1842** to a second surface **1860**. The second surface **1860** extends longitudinally relative to the central transverse axis **220** from the first surface **1858** to a third surface **1862**. Furthermore, the second surface **1860** includes a hemispherical surface **1864**. As will be discussed in further detail below, the hemispherical surface **1864** is adapted to receive the hemispherical member **1828** of the coupler **1802**. The third surface **1862** extends longitudinally relative to the central vertical axis from the second surface **1860** to the surface **1846** of the second wall **1842**. Furthermore, the first surface **1858** of the hemispherical receptacle **1834** abuts the fourth surface **1856** of the post member **1832**.

(88) FIGS. **19** and **20** depict the coupler **1802** coupled to the safety arm receiver **1803**. When the safety arm receiver **1803** is coupled to coupler **1802**, the second surface **1850**, rounded surface **1852**, and the third surface **1854** of the safety arm receiver **1803** nest within the opening **1830** of the coupler **1802**. In this position, second surface **1850** of the safety arm receiver **1803** abuts the fourth surface **1814** of the coupler **1802** and the third surface **1854** of the safety arm receiver **1803** abuts the sixth surface **1818** of the coupler **1802**. Furthermore, when the coupler **1802** is coupled to the safety arm receiver **1803**, the hemispherical member **1828** of the coupler **1802** nests within and abuts the hemispherical surface **1864** of the safety arm receiver **1803**.

(89) FIG. **21** depicts a safety arm **124** with the coupler **1802** contacting the object **1002**. When the vehicle **100** moves in the direction of arrow **L** and the object **1002** contacts the safety arm **124**, a force is applied to the safety arm **124** causing the safety arm **124** rotate in the direction of arrow **M**. When the safety arm **124** is in the position depicted in FIG. **21** and the vehicle **100** continues to move in the direction of arrow **L**, the safety arm **124** moves in the direction of arrow **N** and detaches or decouples from safety arm receiver **1803**. When decoupled, the safety arm **124** and the coupler **1802** fall away from the vehicle **100** without damaging the vehicle **100**. In this

embodiment, the safety arm **124** may be undamaged and the safety arm **124** may be reconnected to the safety arm receiver **1803**.

(90) FIG. **22** depicts a first safety arm **124a**, a second safety arm **124b**, the bracket **128**, and the bracket support **130** being removed from the vehicle **100**. In order to remove the first safety arm **124a** and the second safety arm **124b**, the first safety arm **124a** and the second safety arm **124b** are moved in the direction of arrow O. In order to remove the bracket **128** and the bracket support **130** from the bracket holder **126**, first a first door **2202** and a second door **2204** are opened and then the bracket **128** and the bracket support **130** are moved in the direction of arrow P and removed from the bracket holder **126**.

(91) FIG. **23** depicts the bracket **128**. The bracket **128** includes a first side **128A**, a second side **128B** opposite the first side **128A**, a first safety arm holder **2302** and a second safety arm holder **2304**. The first safety arm holder **2302** extends generally parallel to a first horizontal axis **2306** and the second safety arm holder **2304** extends generally parallel to a second horizontal axis **2308**. The first horizontal axis **2306** and the second horizontal axis **2308** intersect a central vertical axis **2309** at right angles. Stated another way, the first horizontal axis **2306** and the second horizontal axis **2308** are perpendicular to the central vertical axis **2309**. The first safety arm holder **2302** extends generally perpendicular to the second safety arm holder **2304**.

(92) The bracket **128** is octagonal in shape. A first surface **2310** of the first safety arm holder **2302** defines a first side **2312** of the bracket **128**. A second surface **2314** opposite the first surface **2310** defines a second side **2316** of the first safety arm holder **2302**. The first side **2312** of the bracket **128** is opposite the second side **2316** of the bracket **128**. A semi-circular surface **2318** extends from the first surface **2310** to the second surface **2314** of the first safety arm holder **2302** and defines a channel **2320**. As will be discussed in further detail below, the channel **2320** may retain a safety arm **124**.

(93) The first surface **2310** and the second surface **2314** extend longitudinally relative to the central vertical axis to a third surface **2322** and a fourth surface **2324** of the first safety arm holder **2302**. The third surface **2322** extends to fifth surface **2326** of the first safety arm holder **2302** and the fourth surface **2324** extends to a sixth surface **2328** of the first safety arm holder **2302**. The fifth surface **2326** and the sixth surface **2328** define an opening **2303** of the first safety arm holder **2302**. The fifth surface **2326** and the sixth surface **2328** extend to the semi-circular surface **2318**. The bracket **128** further includes a base wall **2311**. The base wall **2311** includes a first surface **2332**. A seventh surface **2330** of the first safety arm holder **2302** extends longitudinally relative to the central vertical axis **2309** from the first surface to the third surface **2322**. An eighth surface **2334** extends from the first surface **2332** to the fourth surface **2324**. The first surface **2310**, the second surface **2314**, the fifth surface **2326**, the sixth surface **2328**, the seventh surface **2330**, and the eighth surface **2334** extend generally perpendicular to the first horizontal axis **2306**. The third surface **2322** and the fourth surface **2324** extend generally perpendicular to the first horizontal axis **2306**.

(94) The second safety arm holder **2304** includes a first portion **2336**, a structurally identical second portion **2338**, and an opening **2307** between the first portion **2336** and the second portion **2338**. The first portion **2336** includes a first surface **2340**. The first surface **2340** defines a third side **2342** of the bracket **128** and the second portion **2338** includes a first surface **2343** that define a fourth side **2345** of the bracket **128**.

(95) The third side **2342** and the fourth side **2345** are approximately 90° offset from the first side **2312** and the second side **2316** relative to the central vertical axis **2309**. The first portion **2336** further includes a second surface **2344** opposite the first surface **2340**. A semi-circular surface **2346** extends from the first surface **2340** to the second surface **2344** of the second safety arm holder **2304** and defines a channel **2348**. The channel **2348** extends from the first surface **2340** of the first portion **2336** to the first surface **2343** of the second portion **2338**. As will be discussed in further detail below, the channel **2348** may retain a safety arm **124**. The semi-circular surface **2346** is

vertically above the third surface **2322** and the fourth surface **2324** of the first safety arm holder **2302** relative to the central vertical axis **2309**.

(96) The first surface **2340** of the first portion **2336** and the second surface **2344** of the first portion **2336** extend longitudinally relative to the central vertical axis **2309** to a third surface **2350** and a fourth surface **2352** of the second safety arm holder **2304**. Furthermore, the first surface **2343** of the second portion **2338** and the second surface **2347** of the second portion **2338** extend to the third surface **2350** and the fourth surface **2352**. The third surface **2350** extends to fifth surface **2354** of the second safety arm holder **2304** and the fourth surface **2352** surface extends to a sixth surface **2356** of the second safety arm holder **2304**. The fifth surface **2354** and the sixth surface **2356** extend to the semi-circular surface **2346**.

(97) A seventh surface **2358** of the second safety arm holder **2304** extends longitudinally relative to the central vertical axis **2309** from the first surface **2332** of the bracket **128** to the third surface **2350**. An eighth surface **2360** that is opposite the seventh surface **2358** extends from the first surface **2332** of the bracket **128** to the fourth surface **2352**. The first surface **2340**, the second surface **2344**, the fifth surface **2354**, the sixth surface **2356**, the seventh surface **2358**, and the eighth surface **2360** extend generally perpendicular to the central vertical axis **2309**. The third surface **2322** and the fourth surface **2324** extend generally parallel to the second horizontal axis **2308**.

(98) FIG. **24** depicts the second side **128B** of the bracket **128**. The bracket **128** includes a first bracket support holder **2402** and a structurally identical second bracket support holder **2404** that extend longitudinally with respect to the central vertical axis **2309** from a surface **2406** of the bracket **128**. Accordingly, for the sake of brevity, only the first bracket support holder **2402** will be discussed in further detail below.

(99) The first bracket support holder **2402** includes a first surface **2408**, a second surface **2410**, a third surface **2412**, and a fourth surface **2414**. The first surface **2408** extends at an approximately 45° angle from the surface **2406** to the second surface **2410** and extends between the third surface **2412** and the fourth surface **2414**. The second surface **2410** extends at an approximately 45° angle from the surface **2406** to the first surface **2408**. The second surface **2410** faces a side of the bracket **128** that is not defined by the first safety arm holder **2302** or the second safety arm holder **2304**. The second surface **2410** includes an annular edge **2416** that defines a bracket support receptacle **2418**. As will be discussed in further detail below, the bracket support receptacle **2418** may be adapted to receive a bracket support **130**. The third surface **2412** and the fourth surface **2414** extend longitudinally relative to the central vertical axis **2309** to the first surface **2408** and the second surface **2410**.

(100) FIG. **25A** depicts the first safety arm **124a** coupled to the bracket **128**. In order to couple the safety arm **124a** to the bracket **128**, the first safety arm **124a** is moved in the direction of arrow Q. A safety arm **124** includes a rounded surface **2502** that extends between the first side **902** and the second side **904** of the safety arm **124**. A width between opposing sides of the rounded surface **2502** is greater than a width between the first side **902** and the second side **904**. Furthermore, the width between opposing sides of the rounded surface **2502** is greater than a width of the opening **2303** and the opening **2307**, whereas the width between the first side **902** and the second side **904** is less than a width of the opening **2303** and the opening **2307**. Accordingly, when inserting a safety arm **124** into the bracket **128**, the safety arm is oriented so that the first side **902** and the second side **904** are parallel to the central vertical axis **2309** allow a safety arm **124** to nest within the first safety arm holder **2302** or the second safety arm holder **2304**. The safety arm **124** may be rotated in the direction of arrow S so that the first side **902** and the second side **904** are perpendicular to the central vertical axis **2309** so that the first safety arm holder **2302** and the second safety arm holder **2304** retain a safety arm **124**.

(101) Furthermore, a safety arm **124** has a midpoint between the first end **360** and the second end **362**. The safety arm **124a** may be moved in either direction of arrow R in order to align the

midpoint of the safety arm **124a** with a center of the bracket **128**. As depicted in FIG. 25A, when coupling the safety arm **124a** to the bracket **128**, the light source **345** may be adjacent to the semi-circular surface **2318**.

(102) FIG. 25C depicts the first safety arm **124a** and the second safety arm **124b** coupled to the bracket **128**. The second safety arm **124b** is coupled to the bracket **128** in an identical manner as the first safety arm **124a**. The safety arm **124b** may be moved until the midpoint of the safety arm **124b** aligns with the midpoint of the first safety arm **124a**. When the first safety arm **124a** and the second safety arm **124b** are coupled to bracket **128** and the midpoints are aligned with a center of the bracket **128**, the first safety arm **124a** and the second safety arm **124b** form an “X” shape.

(103) FIG. 26 depicts the bracket **128** coupled to the bracket support **130**. In this embodiment, the bracket support receptacle **2418** slidably receives the bracket support **130**. The bracket **128**, the bracket support **130**, the first safety arm **124a**, and the second safety arm **124b** form a safety device **2602**. When assembled, the safety device **2602** may stand vertically on a surface **2604**. When standing, the first safety arm **124a**, the second safety arm **124b**, and the bracket support **130** contact the surface **2604** to support the safety device **2602**. While FIG. 26 shows the first end **360** of the first safety arm **124a** and the second safety arm **124b**, contacting the surface **2604**, it is envisioned that the first safety arm **124a** and the second safety arm **124b** may be rotated 180° so that the second end **362** of the first safety arm **124a** and the second safety arm **124b** may contact the surface **2604**.

(104) FIG. 27 depicts the safety device **2602** in operation. When in operation, the safety device **2602** may be positioned adjacent to the vehicle **100**. In one embodiment, when in operation, the safety device **2602** emits light **2702** from the light source **345** of the first safety arm **124a** and the second safety arm **124b**. In another embodiment, when in operation, the safety device **2602** does not emit light.

(105) While FIG. 27 shows light **2702** emitted from only one side of the safety device **2602** it is envisioned, in an embodiment wherein the safety arms **124a** and **124b** include a light source **345** on the first side **902** and the second side **904**, so that the safety device **2602** may emit light **2702** from multiple sides. In this embodiment, the safety device **2602** may illuminate multiple areas.

(106) Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(107) While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present

disclosure.

(108) All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

(109) The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

(110) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

(111) When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply

to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

(112) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

(113) Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

(114) An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

(115) If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

(116) As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

(117) Additionally, any method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

(118) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-

closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures.

(119) In the foregoing description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

(120) Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A vehicle comprising: a first safety arm; a first coupler on the first safety arm and rigidly fixed to the first safety arm, wherein the first coupler couples the first safety arm to the vehicle and wherein the first safety arm is adapted to fall away from the vehicle without damaging the vehicle when an object contacts the first safety arm; and a receiver connected to the vehicle and coupled to the first coupler, wherein a portion of the first coupler is disposed inside of the receiver.
2. The vehicle of claim 1, further comprising: a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer.
3. The vehicle of claim 1, further comprising: a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the cab.
4. The vehicle of claim 1, further comprising: a second safety arm; a second coupler on the second safety arm, wherein the second coupler couples the second safety arm to the vehicle and wherein the second safety arm is adapted to fall away from the vehicle without damaging the vehicle when the object contacts the second safety arm.
5. The vehicle of claim 4, further comprising: a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer and the second safety arm is coupled to the cab.
6. The vehicle of claim 4, further comprising: a trailer; and a cab connected to the trailer, wherein the first safety arm and the second safety arm are coupled to the trailer.
7. The vehicle of claim 6, further comprising: a first side of the trailer; a second side of the trailer that is opposite the first side; and a nozzle assembly on the first side, wherein the first safety arm and the second safety arm are coupled to the first side forward and rearward of the nozzle assembly, respectively.
8. The vehicle of claim 7, further comprising: a hose holder connected to the first side of the trailer adjacent to the nozzle assembly, wherein the first safety arm and the second safety arm are coupled to the hose holder.
9. The vehicle of claim 1, further comprising: a ferromagnetic surface on one of the coupler and the first safety arm; a magnet on the other of the coupler and the first safety arm; wherein the magnet couples the first safety arm to the ferromagnetic surface.
10. The vehicle of claim 1, further comprising: a cab; a trailer connected to the cab; a nozzle assembly on the trailer; and a hose holder connected to the trailer adjacent to the nozzle assembly, wherein the receiver is connected to the hose holder.
11. The vehicle of claim 1, wherein the first coupler is adapted to break without damaging the vehicle.
12. The vehicle of claim 11, wherein the receiver defines a keyhole and the first coupler includes a first portion within the keyhole and a second portion outside of the keyhole, that is adapted to break when the first safety arm contacts the object.
13. The vehicle of claim 1, wherein the first coupler is adapted to fall away from the receiver with the first safety arm without damaging the vehicle.
14. The vehicle of claim 13, wherein the receiver includes a triangular arm receiver and the first coupler rests within the triangular arm receiver.

15. The vehicle of claim 13, wherein the receiver includes a hemispherical surface and the first coupler includes a hemispherical member that abuts the hemispherical surface of the receiver.
 16. The vehicle of claim 1, wherein the first safety arm includes a light source.
 17. The vehicle of claim 16, wherein the first safety arm includes a battery in electrical communication with the light source.
 18. The vehicle of claim 16, further comprising: a power source; and a power outlet in electrical communication with the power source and connected to a plug of the first safety arm, wherein the plug is in electrical communication with the light source.
 19. The vehicle of claim 1, wherein the first safety arm is coupled to the vehicle without any support structure depending from the first safety arm.
 20. A method for detaching a safety arm from a vehicle comprising: providing a coupler on a safety arm, wherein the coupler is rigidly fixed to the safety arm; attaching the safety arm to a vehicle with the coupler; housing a portion of the coupler inside of a receiver that is connected to the vehicle; applying a force to the safety arm; and detaching the safety arm from the vehicle with the applied force without damaging the vehicle.
 21. The method for detaching a safety arm from a vehicle of claim 20, further comprising: illuminating an area adjacent to the vehicle with a light source on the arm while the arm is attached to the vehicle.
 22. The method for detaching a safety arm from a vehicle of claim 20, further comprising: moving the vehicle; contacting an object with the arm while the vehicle is moving, wherein the contact applies the force to the arm; and detaching the arm from the vehicle while the vehicle is moving without damaging the vehicle.
 23. The method for detaching a safety arm from a vehicle of claim 20, further comprising: breaking a coupler of the arm with the applied force, wherein breaking the coupler detaches the arm from the vehicle without damaging the vehicle.
 24. The method for detaching a safety arm from a vehicle of claim 20, wherein the arm detaches from the vehicle by falling away from the vehicle without damaging the vehicle.
 25. A vehicle comprising: a first safety arm; a first coupler the first safety arm, wherein the first coupler couples the first safety arm to the vehicle and wherein the first safety arm is adapted to fall away from the vehicle without damaging the vehicle when an object contacts the first safety arm; a second safety arm; and a second coupler on the second safety arm, wherein the second coupler couples the second safety arm to the vehicle and wherein the second safety arm is adapted to fall away from the vehicle without damaging the vehicle when the object contacts the second safety arm.
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